$\mathrm{pr}_2((1))$

$$= \begin{bmatrix} \mathbf{A}_g & \mathbf{0} \\ \mathbf{B}_f \begin{bmatrix} \mathbf{C}_g \\ \mathbf{0} \end{bmatrix} & \mathbf{A}_f \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{B}_f^{[:,i+1:i+k]} \end{bmatrix} \mathbf{F} \begin{bmatrix} \mathbf{D}_f^{[:,1:i]}\mathbf{C}_g & \mathbf{C}_f \end{bmatrix}^{[j+1:j+k,:]}$$

$$= \begin{bmatrix} \mathbf{A}_g & \mathbf{0} \\ \mathbf{B}_f^{[:,1:i]}\mathbf{C}_g & \mathbf{A}_f \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{B}_f^{[:,i+1:i+k]} \end{bmatrix} \mathbf{F} \begin{bmatrix} \mathbf{D}_f^{[:,1:i]}\mathbf{C}_g & \mathbf{C}_f \end{bmatrix}^{[j+1:j+k,:]}$$

$$= \begin{bmatrix} \mathbf{A}_g & \mathbf{0} \\ \mathbf{B}_f^{[:,1:i]}\mathbf{C}_g & \mathbf{A}_f \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{B}_f^{[:,i+1:i+k]} \end{bmatrix} \mathbf{F}' \begin{bmatrix} \mathbf{D}_f^{[j+1:j+k,1:i]}\mathbf{C}_g & \mathbf{C}_f^{[j+1:j+k,:]} \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{A}_g & \mathbf{0} \\ \mathbf{B}_f^{[:,1:i]}\mathbf{C}_g + \mathbf{B}_f^{[:,i+1:i+k]}\mathbf{F}'\mathbf{D}_f^{[j+1:j+k,1:i]}\mathbf{C}_g & \mathbf{A}_f + \mathbf{B}_f^{[:,i+1:i+k]}\mathbf{F}'\mathbf{C}_f^{[j+1:j+k,:]} \end{bmatrix},$$